Beyond $\mathcal{ALC}_{\mathsf{reg}}$: Exploring Non-Regular Extensions of PDL with DL Features

4th of September, DL Workshop 2023 & 22nd of September, JELIA 2023

Bartosz "Bart" Bednarczyk

With special thanks to Reijo Jaakkola, Witek Charatonik, and Sebastian Rudolph for all their support.

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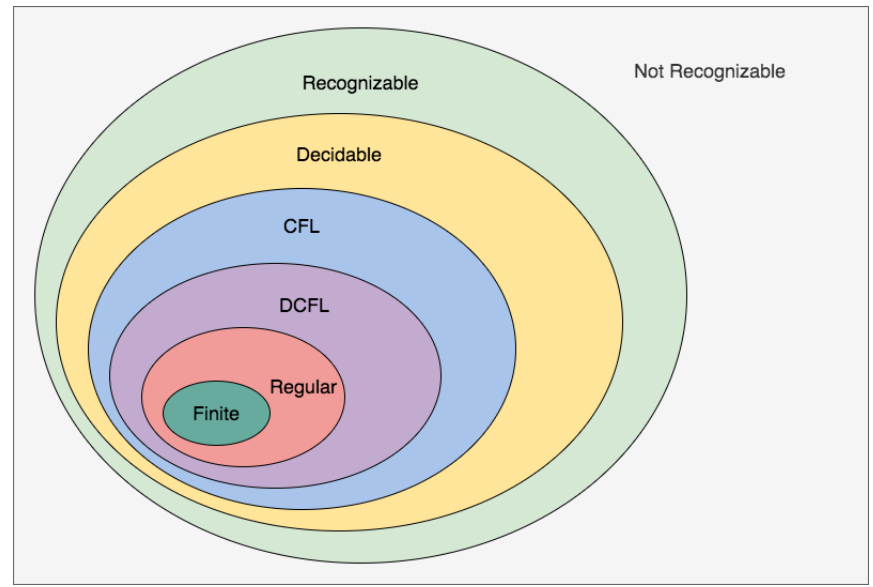

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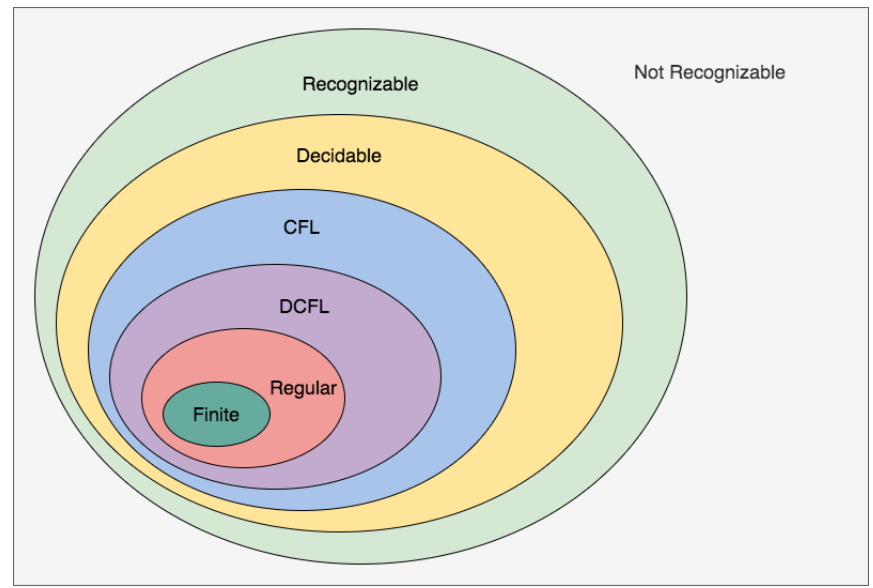

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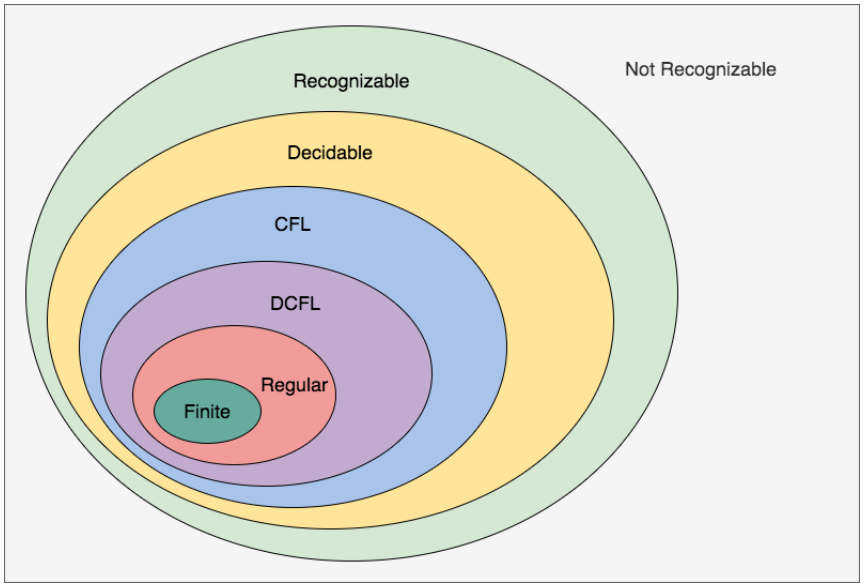

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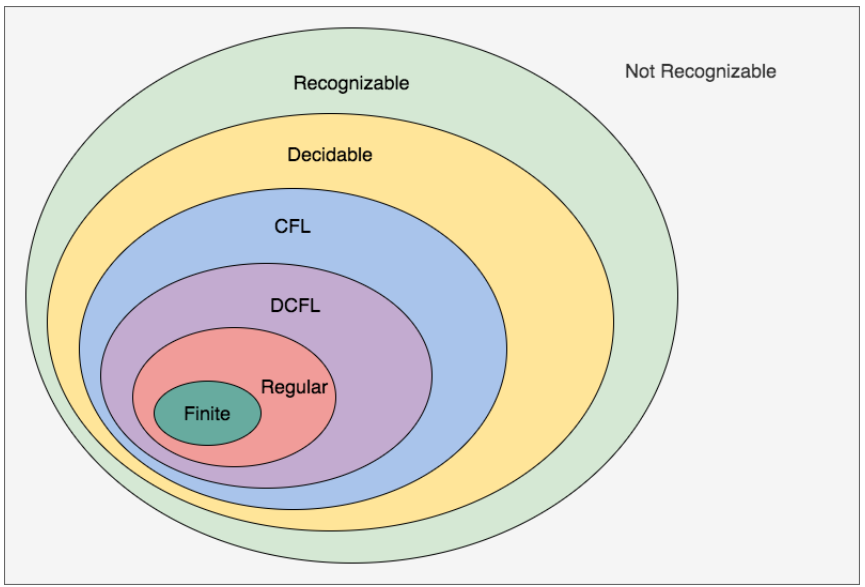

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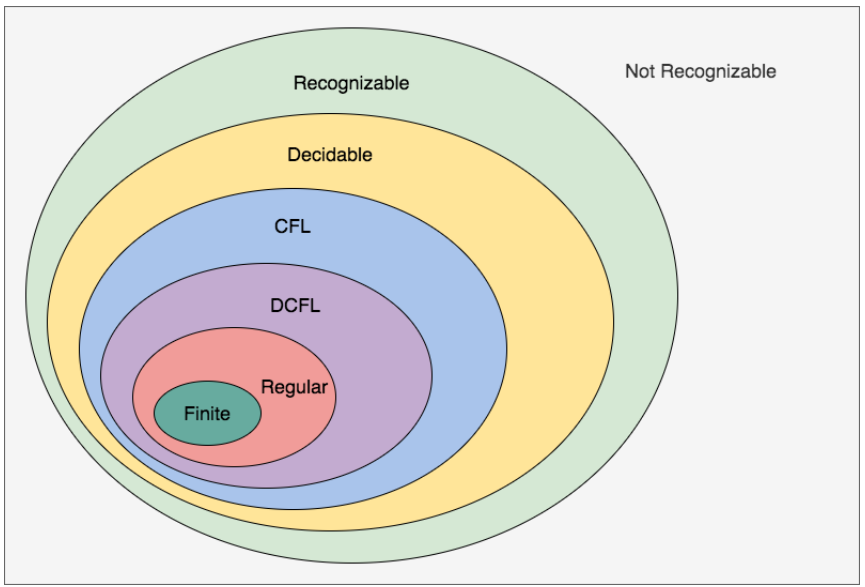

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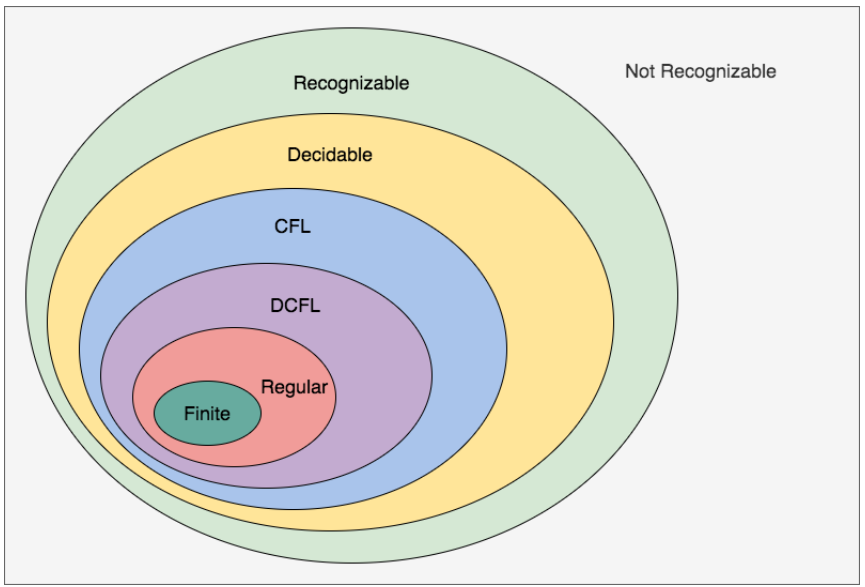

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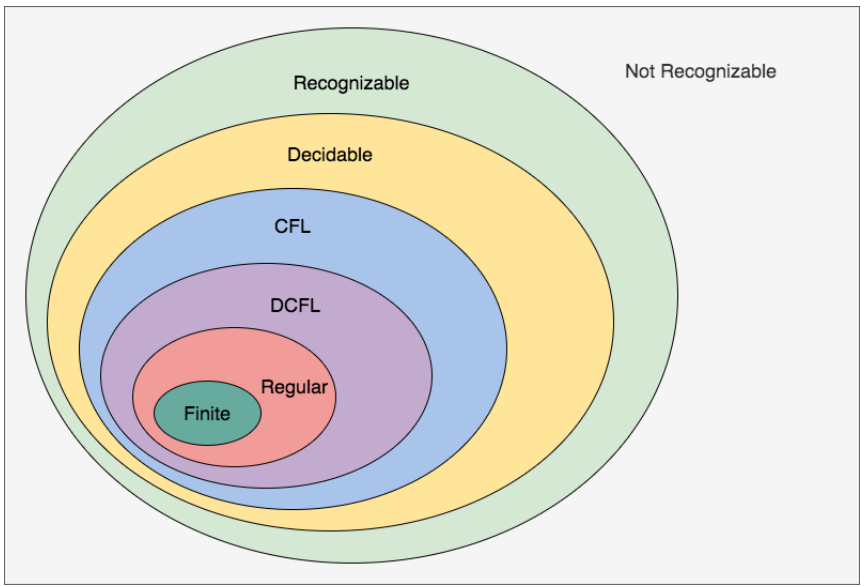

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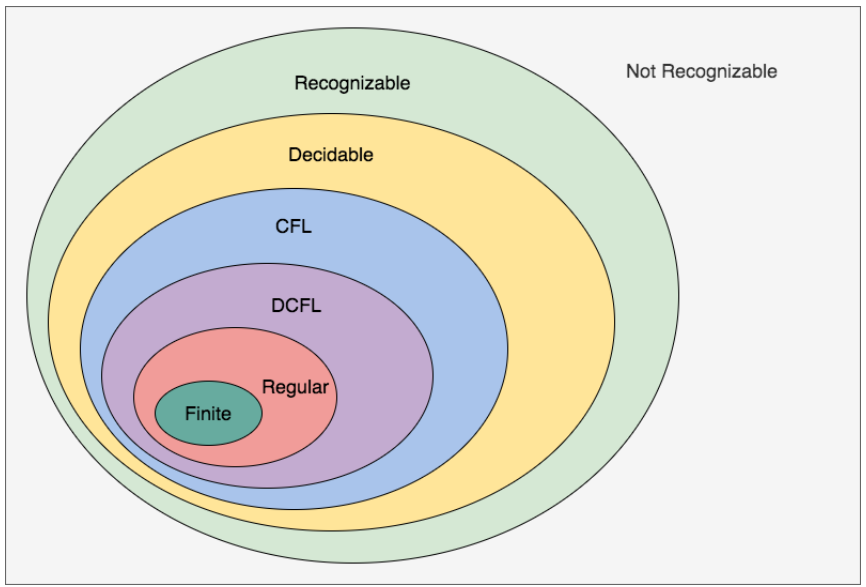

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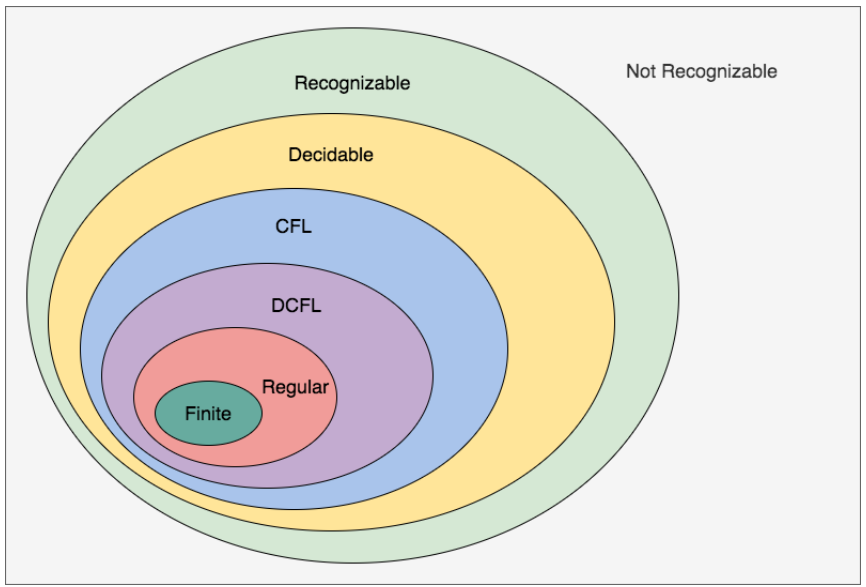
$\mathcal{ALC}$+Visibly Pushdown Languages is decidable.



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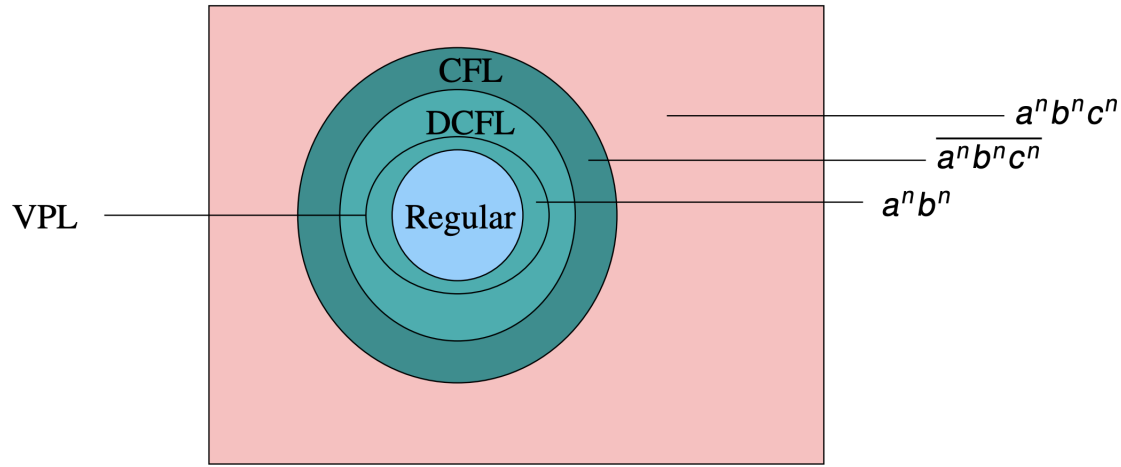

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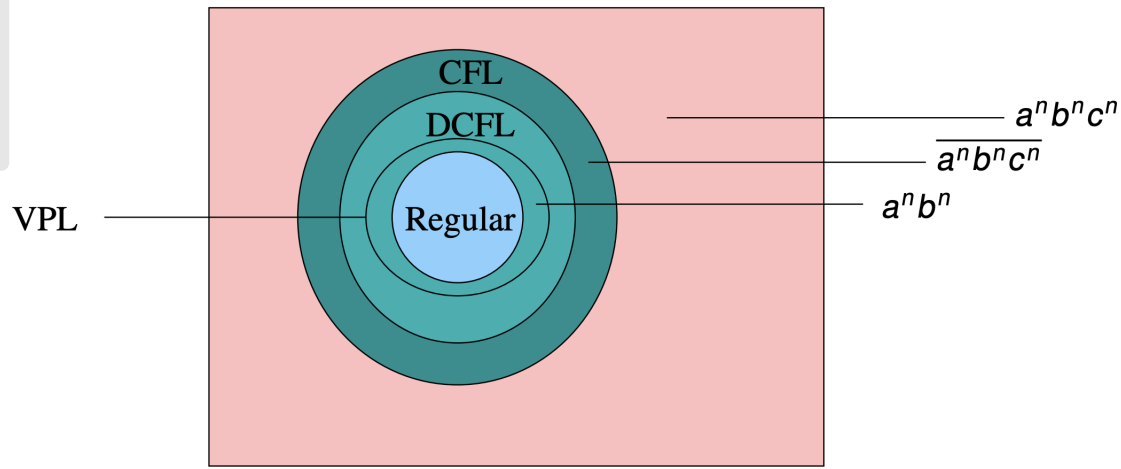

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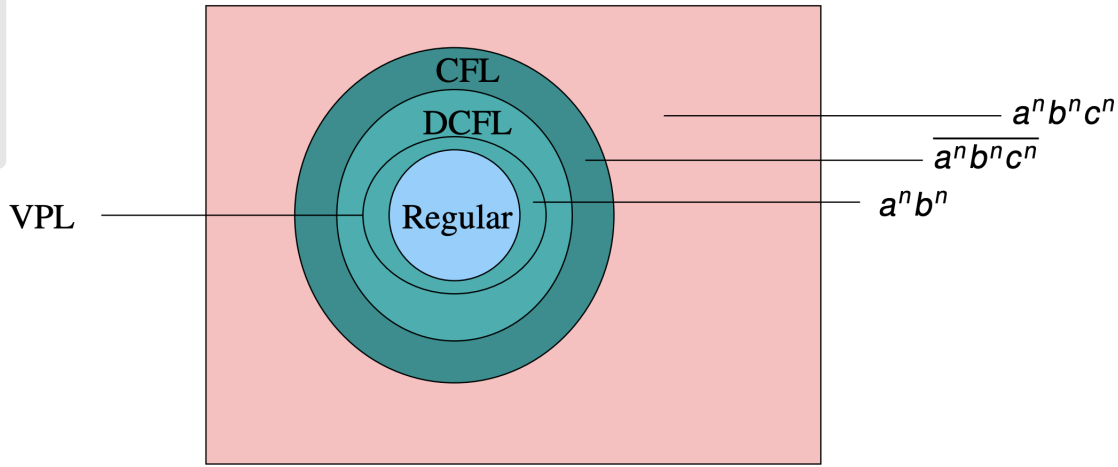

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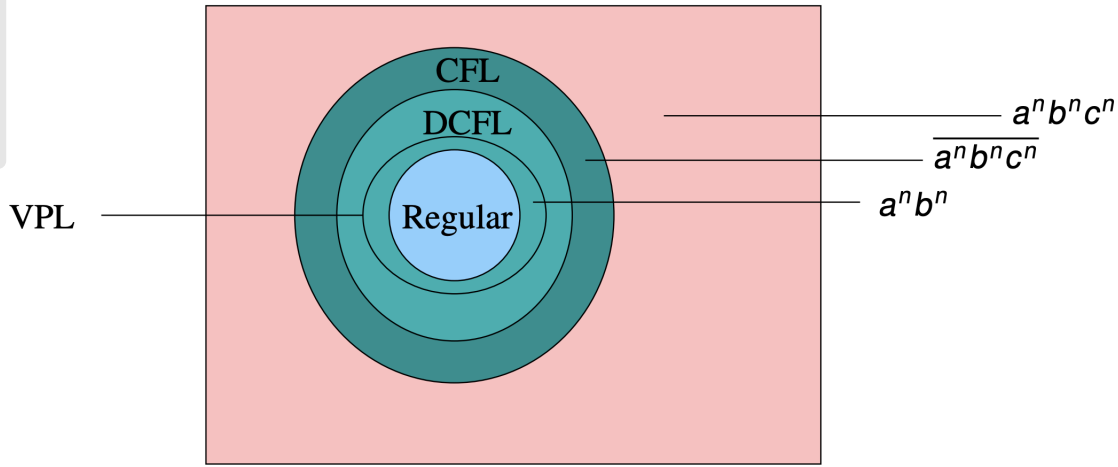

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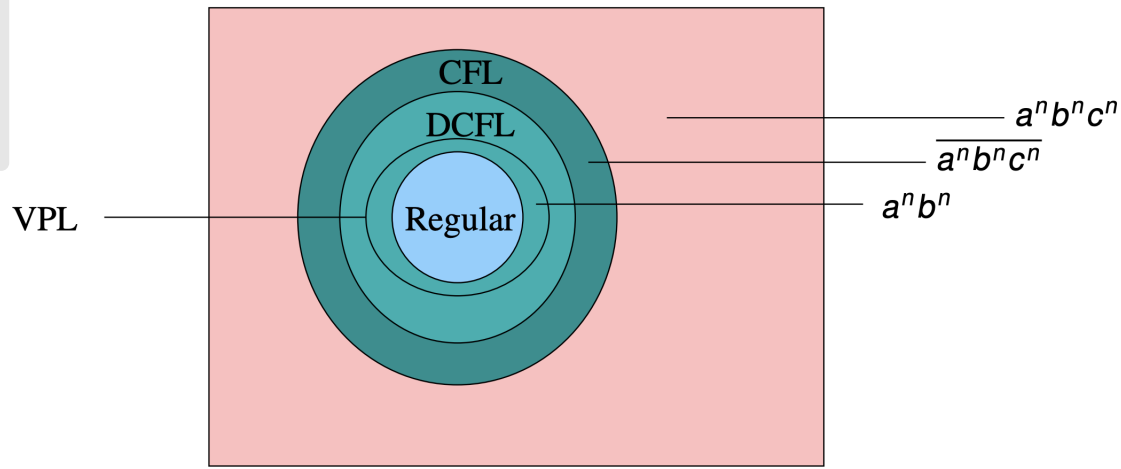

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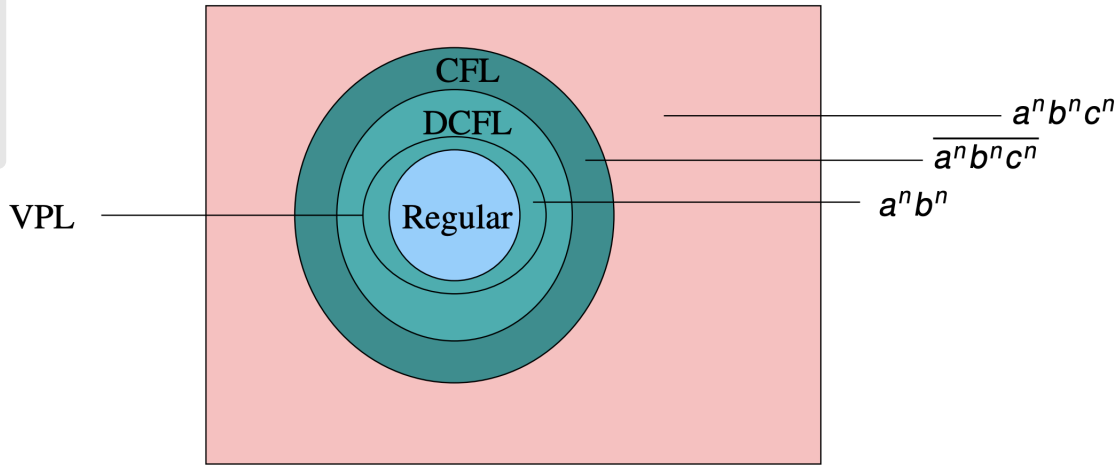

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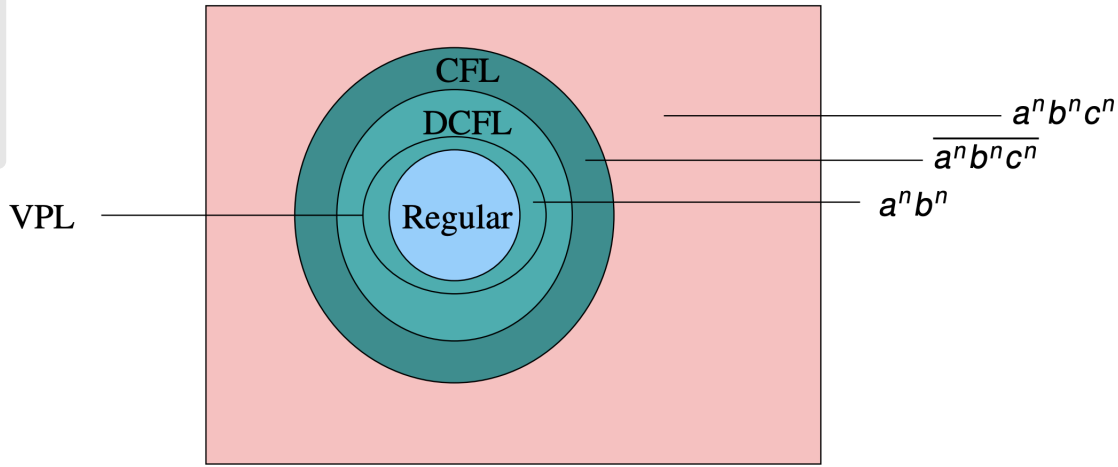

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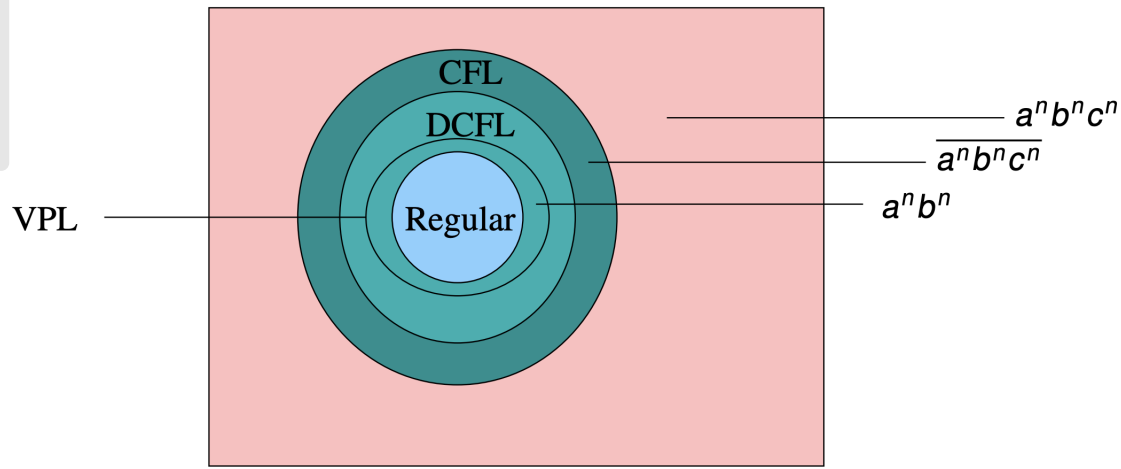

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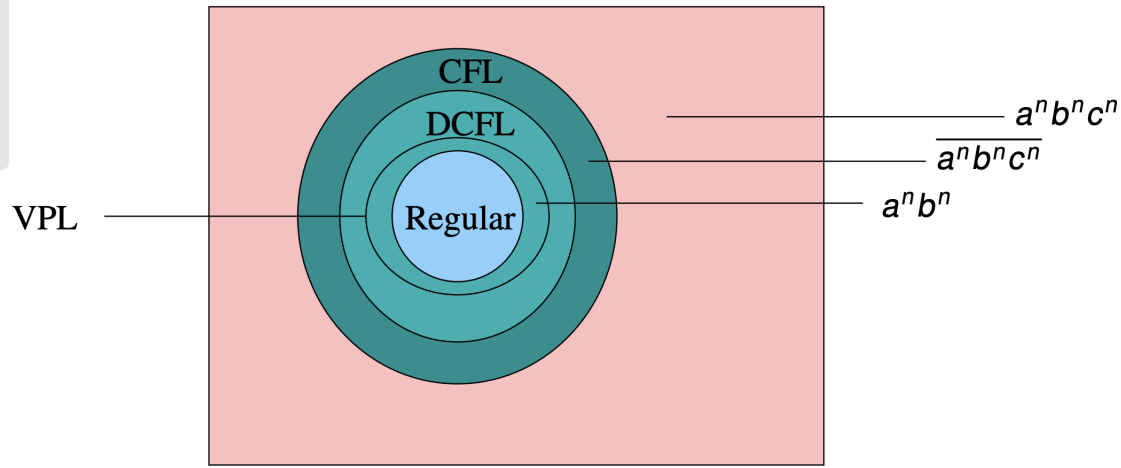

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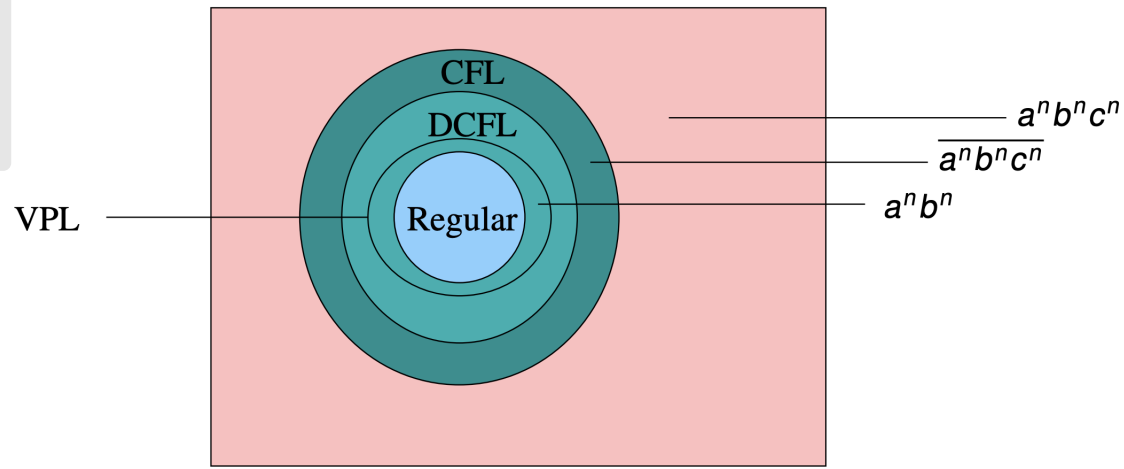

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	Decision problems for automata		
	Emptiness	Universality/Equivalence	Inclusion
NFA	NLOGSPACE	PSPACE	PSPACE
PDA	PTIME	Undecidable	Undecidable
DPDA	PTIME	Decidable	Undecidable
VPA	PTIME	EXPTIME	EXPTIME

	Closure under				
	Union	Intersection	Complement	Concat.	Kleene-$*$
Regular	Yes	Yes	Yes	Yes	Yes
CFL	Yes	No	No	Yes	Yes
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VPL	Yes	Yes	Yes	Yes	Yes

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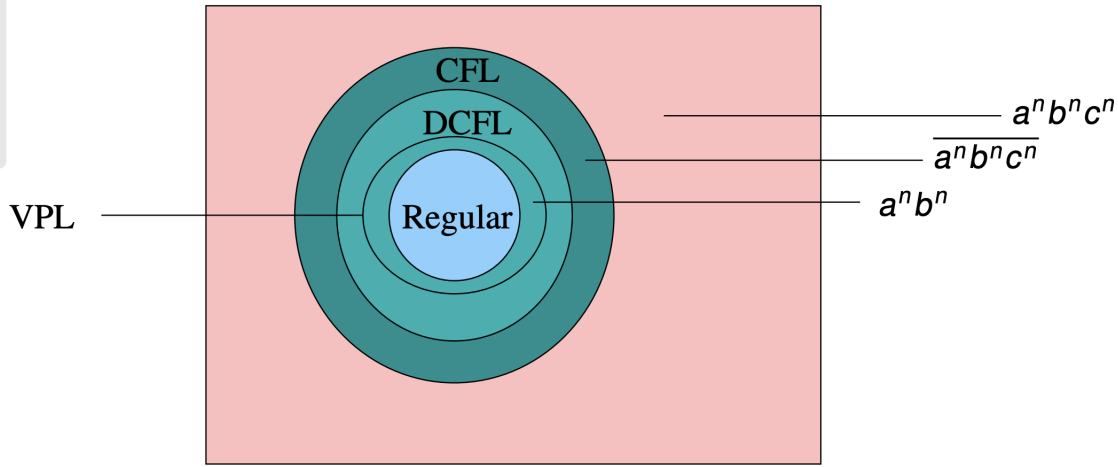

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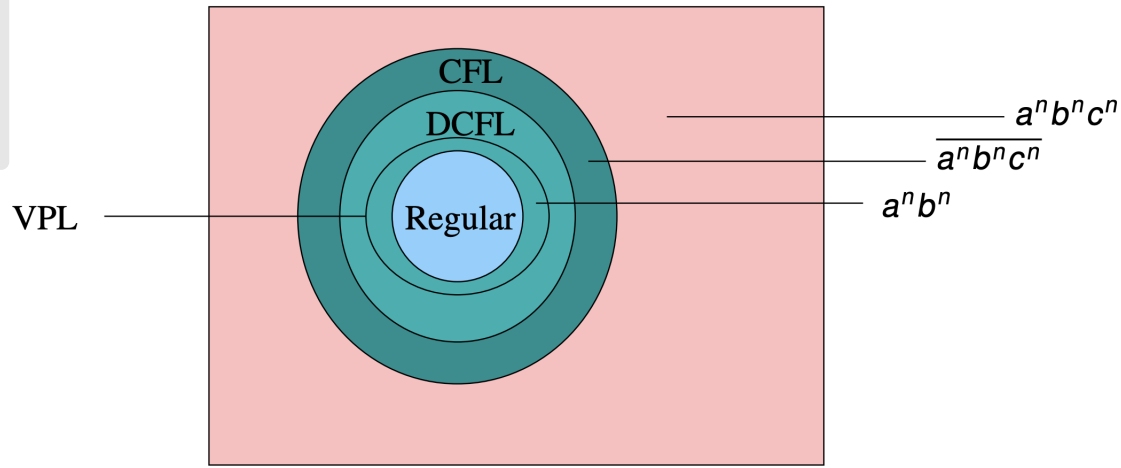

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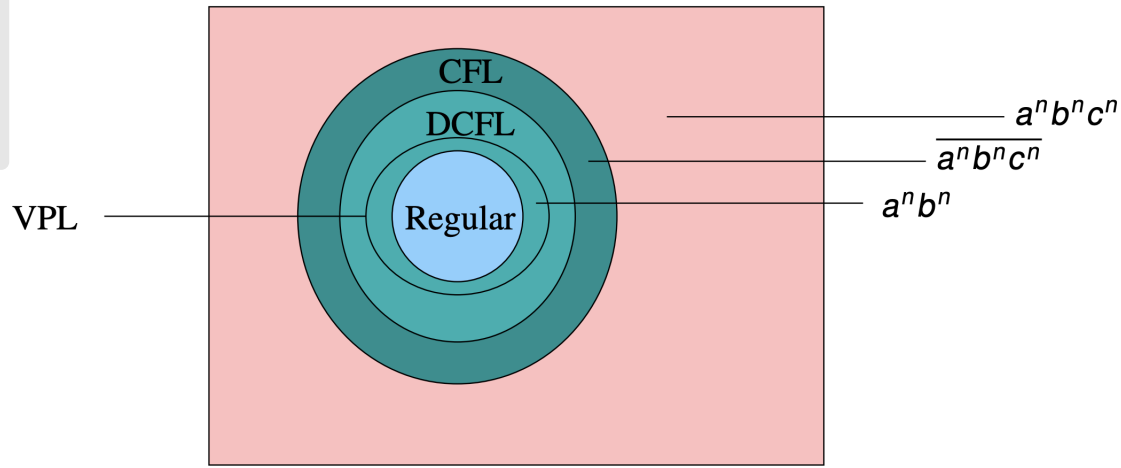

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Why not to employ $\mathbb{VPL}$ in knowledge representation?

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Beyond $\mathcal{ALC}_{\mathsf{reg}}$: Exploring Non-Regular Extensions of PDL with Description Logics Features

Bartosz Bednarczyk[1,2]

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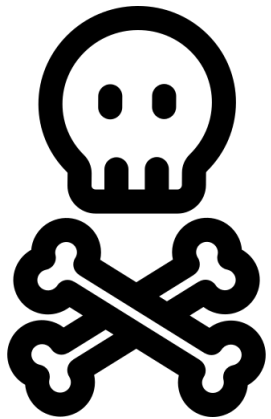

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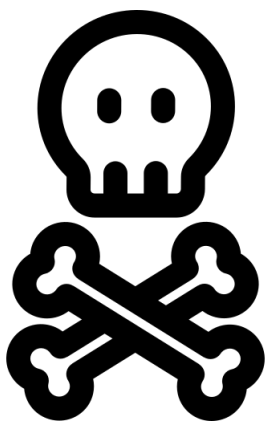
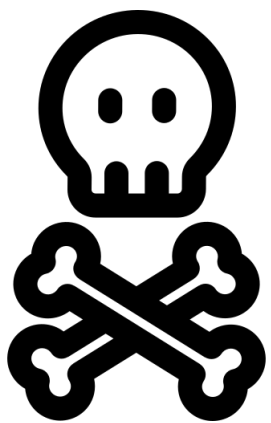
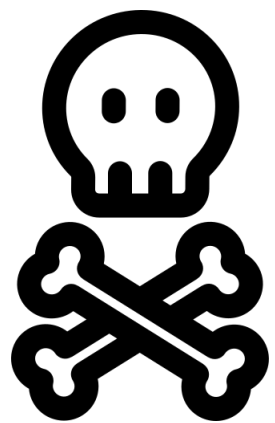

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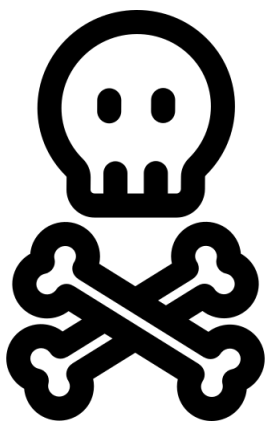
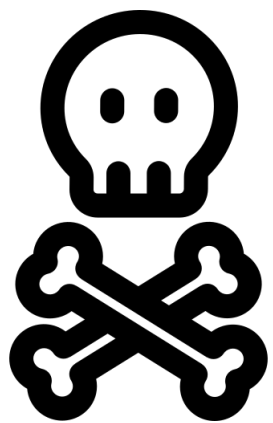


Visibly one counter

Nominals



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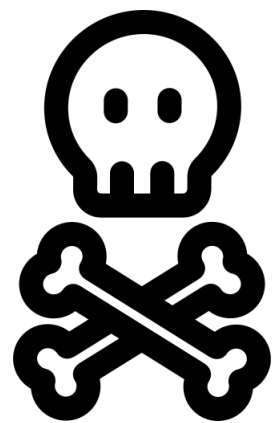

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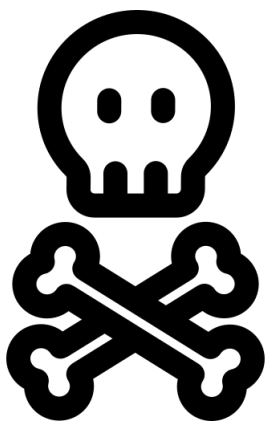

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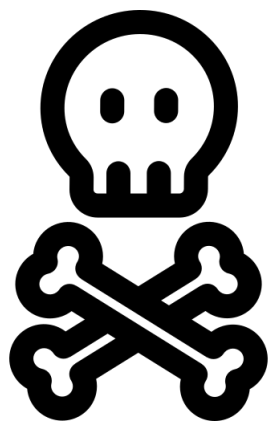

$\mathcal{ALC}_{\mathsf{reg}}^{r\#s\#}$

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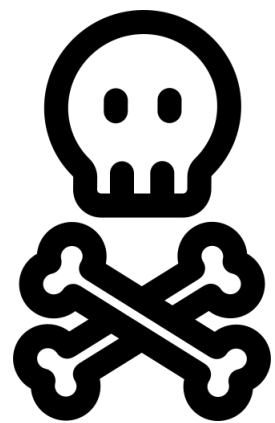

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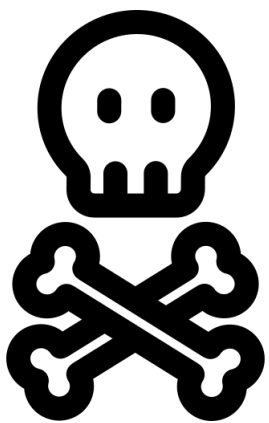

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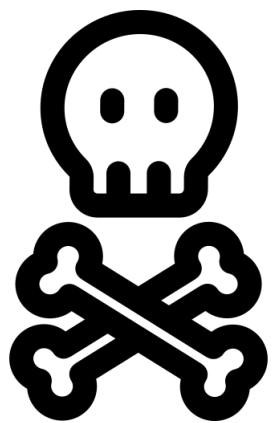

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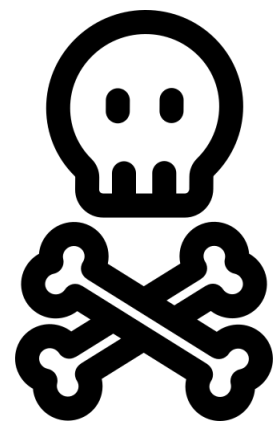

$\mathcal{ALC}$-TBoxes + CRPQs with $r^{\#}s^{\#}$

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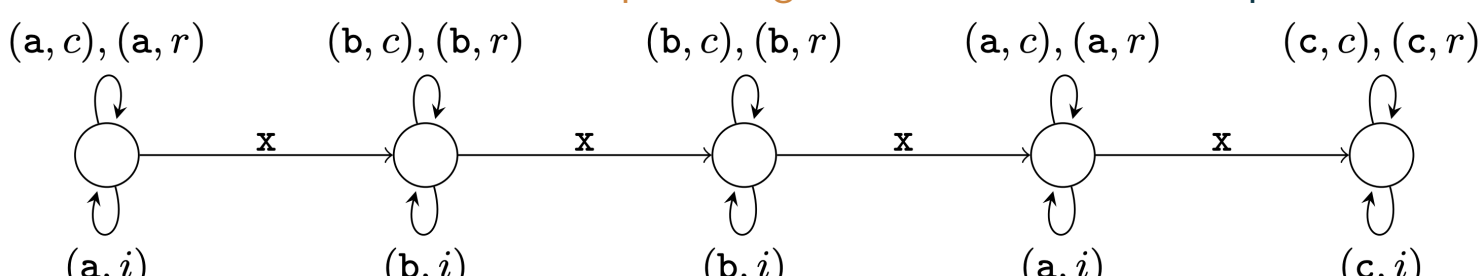

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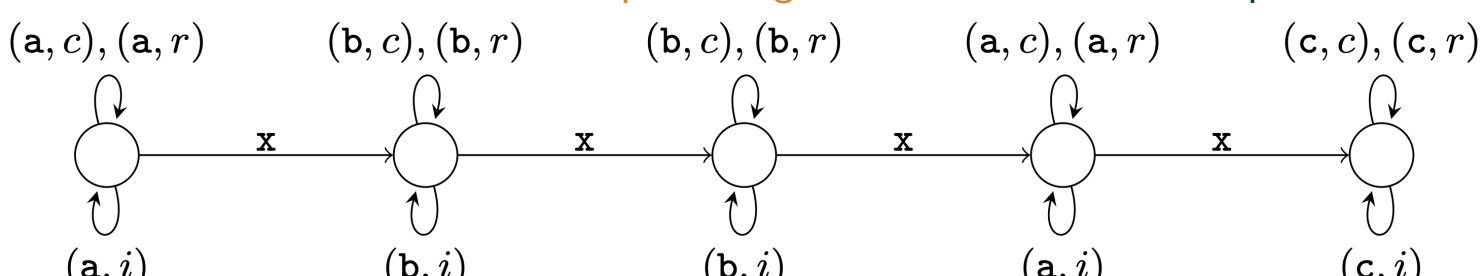

Trick 2: Employ concepts $\forall \hat{\mathcal{A}}_1.\mathrm{OK}_1 \sqcap \forall \hat{\mathcal{C}}_1.\neg \mathrm{OK}_1$ to decorate interpretations with "acceptance" of $\mathcal{A}_1$.

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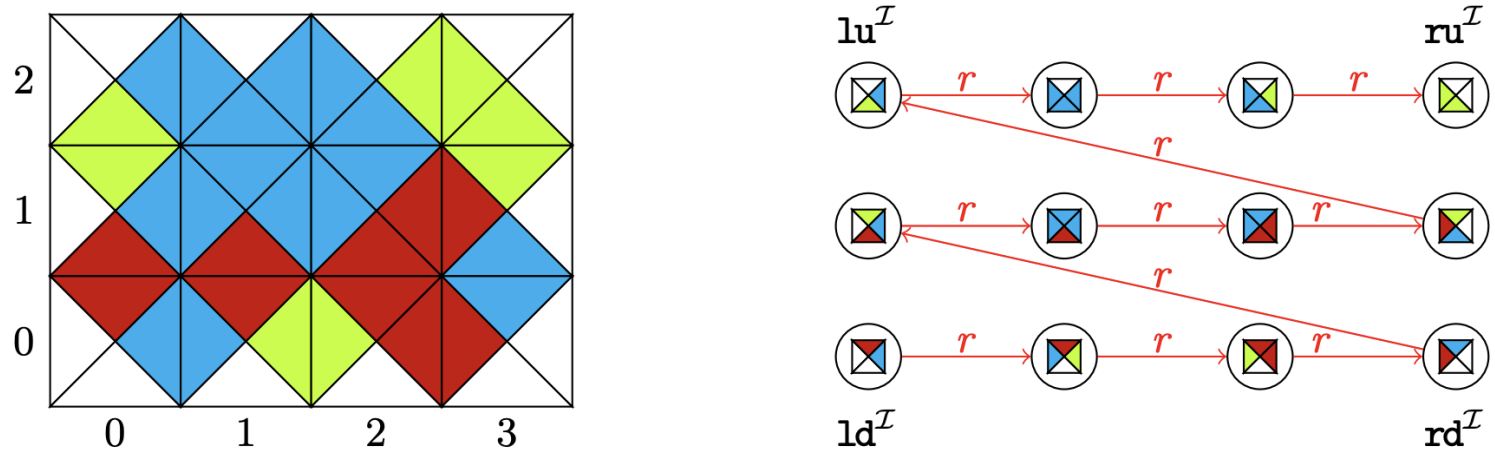

(a) Visualization of ξ.

(b) The encoding of ξ as a $\mathcal{D}$-snake $\mathcal{I}$.

Fig. 1: If $\mathrm{Col} = \{\square, \square, \square, \square\}$ and $\mathrm{T} = \mathrm{Col}^4$, the map $\xi := \{(0,0) \mapsto \boxtimes, (1,0) \mapsto \boxtimes, (2,0) \mapsto \boxtimes, (3,0) \mapsto \boxtimes, (0,1) \mapsto \boxtimes, (1,1) \mapsto \boxtimes, (2,1) \mapsto \boxtimes, (3,1) \mapsto \boxtimes, (0,2) \mapsto \boxtimes, (1,2) \mapsto \boxtimes, (2,2) \mapsto \boxtimes, (3,2) \mapsto \boxtimes\}$ covers $\mathbb{Z}_4 \times \mathbb{Z}_3$.

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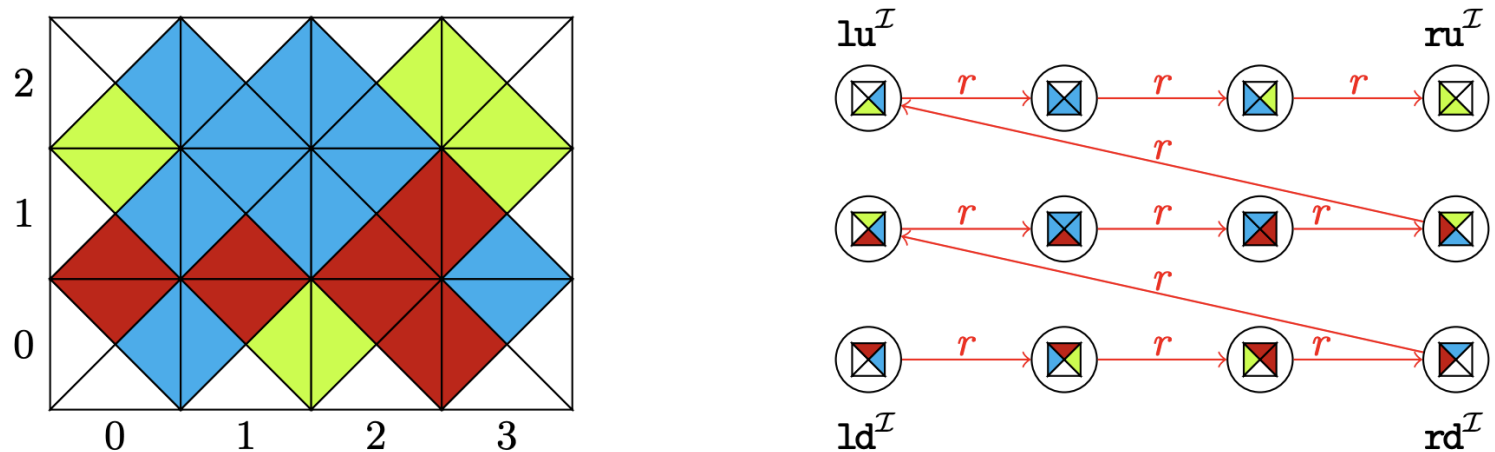

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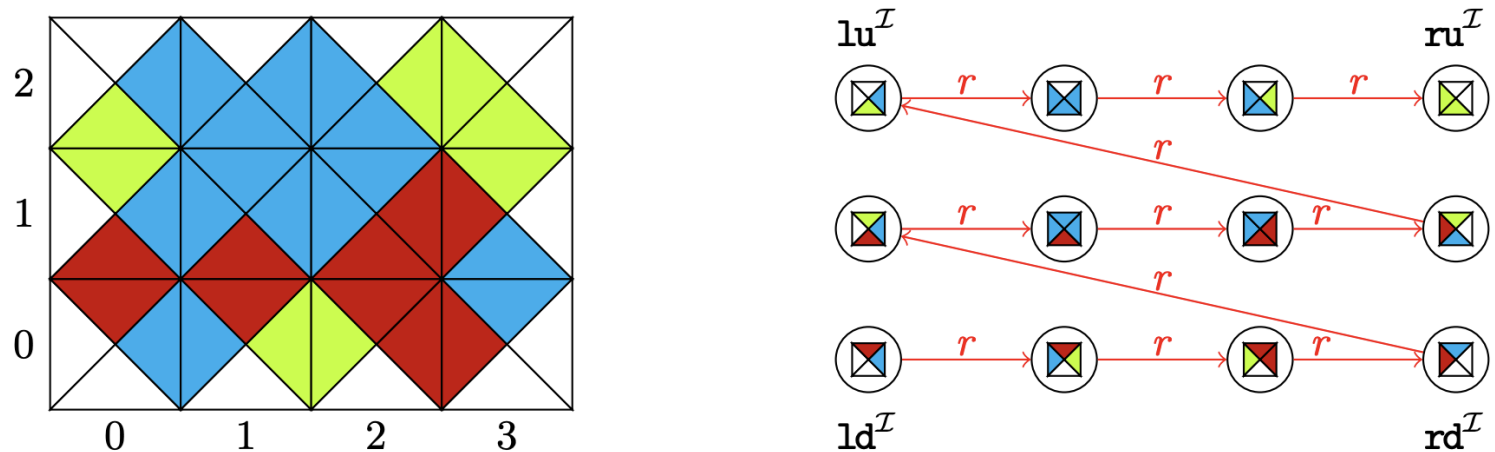

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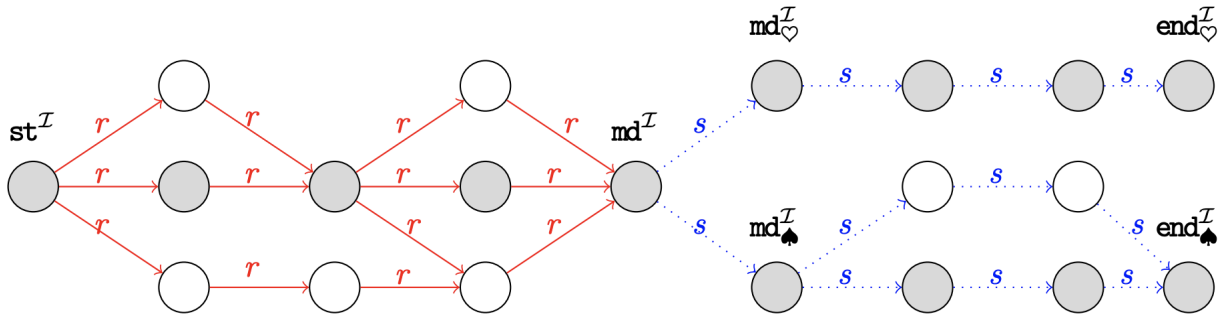

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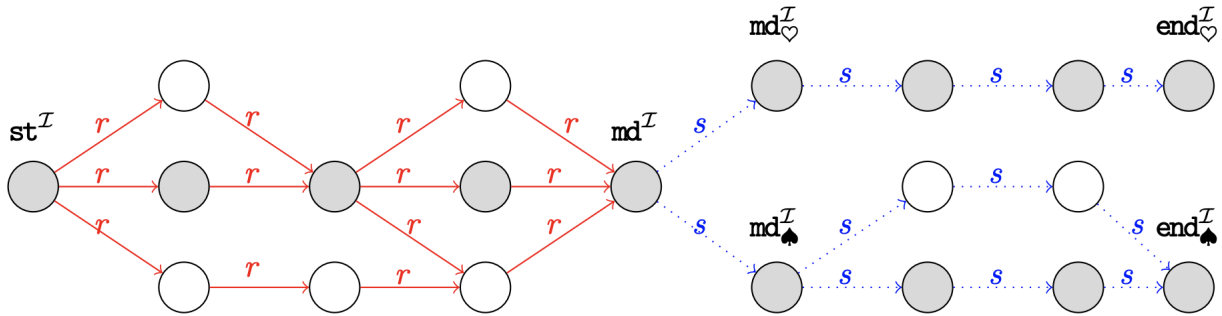

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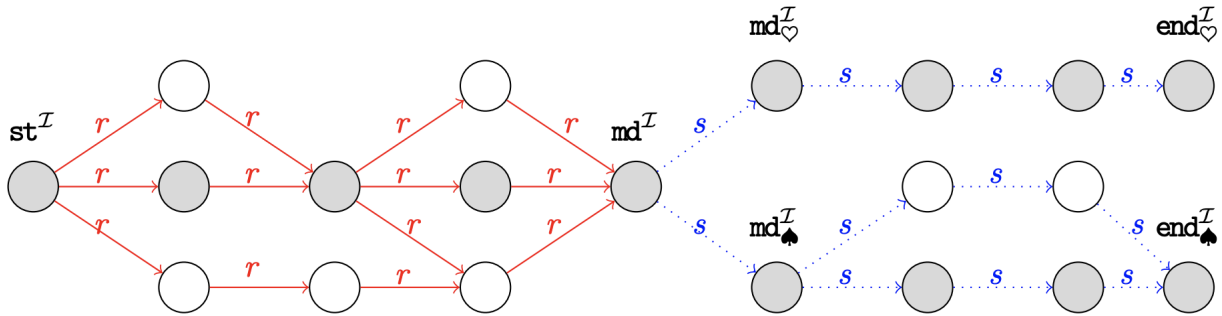

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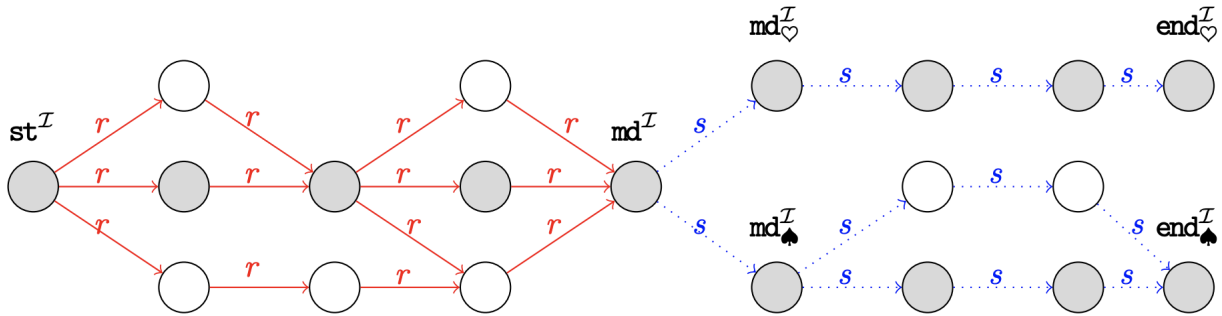

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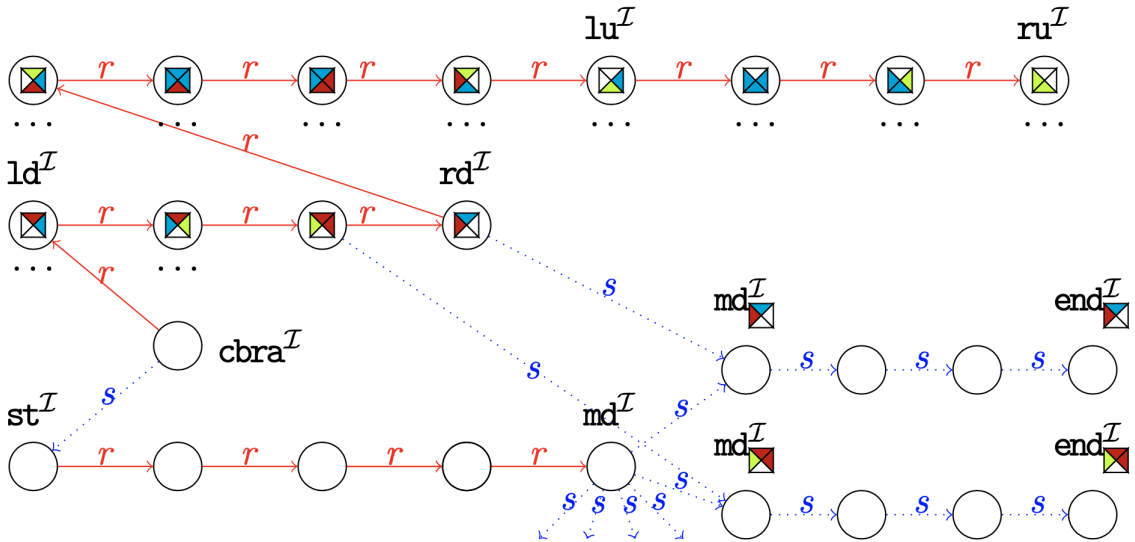

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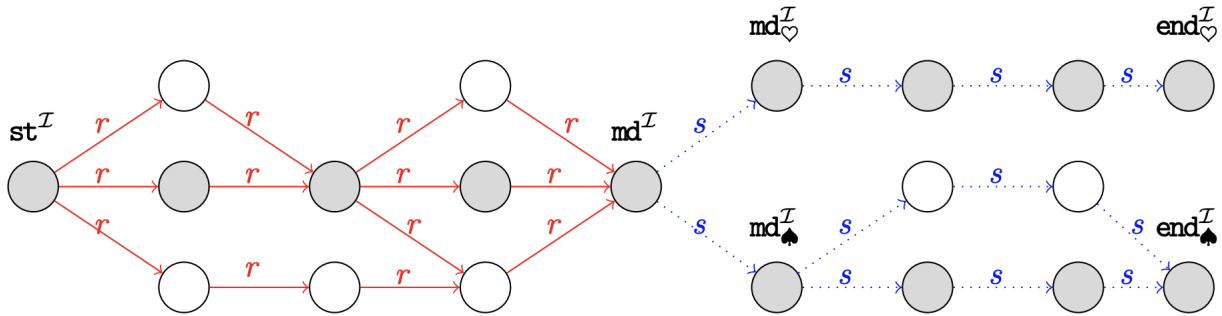

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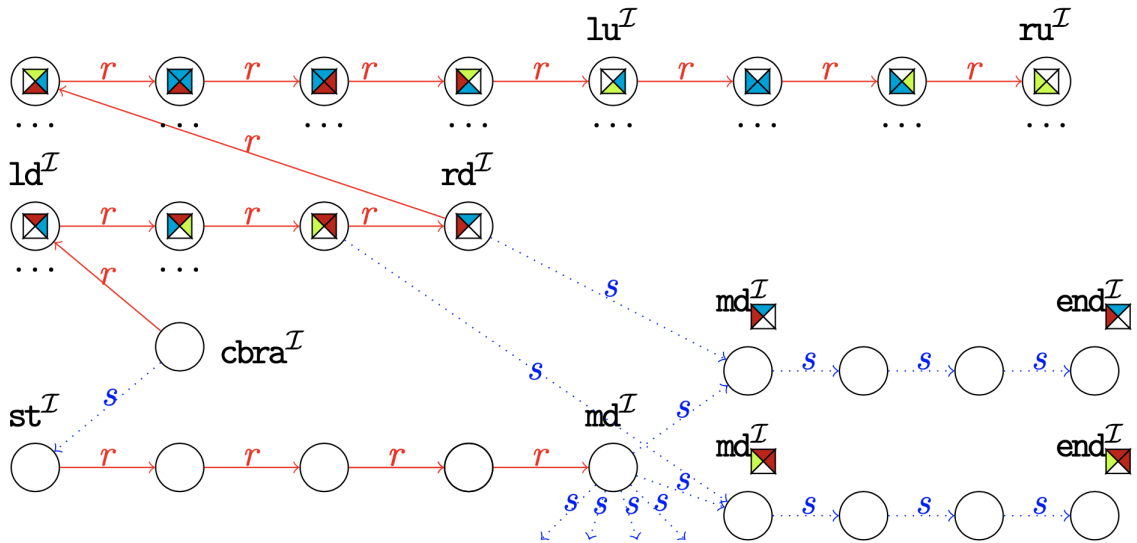
Key property:

An element N steps after d carries a tile t iff d can $r^{\#}s^{\#}$ reach $\mathtt{end}_{\mathrm{t}}$.



Proof sketch: Undecidability of querying $\mathcal{ALC}$-TBoxes with non-regular queries

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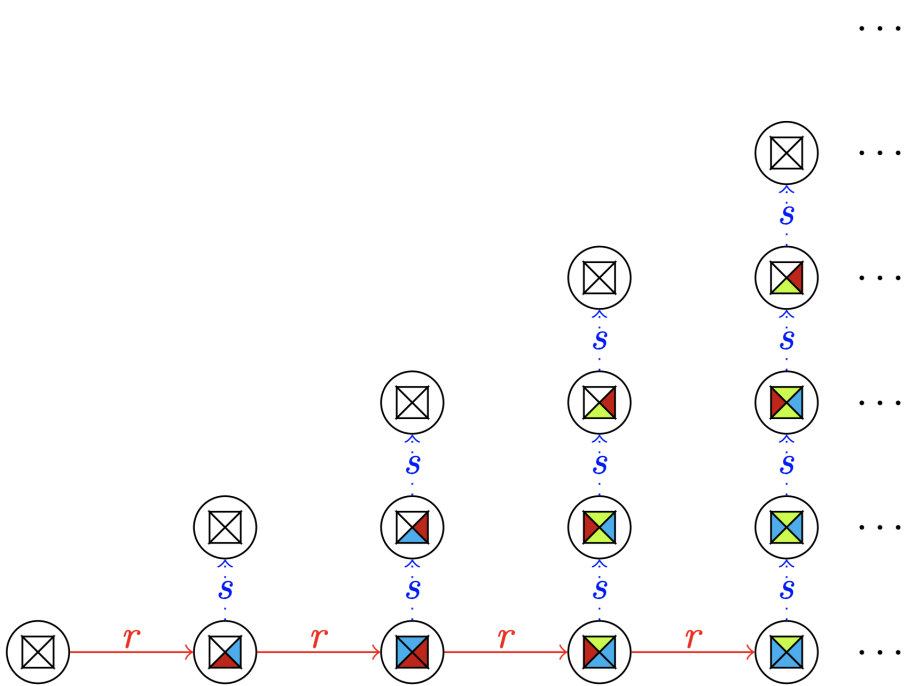

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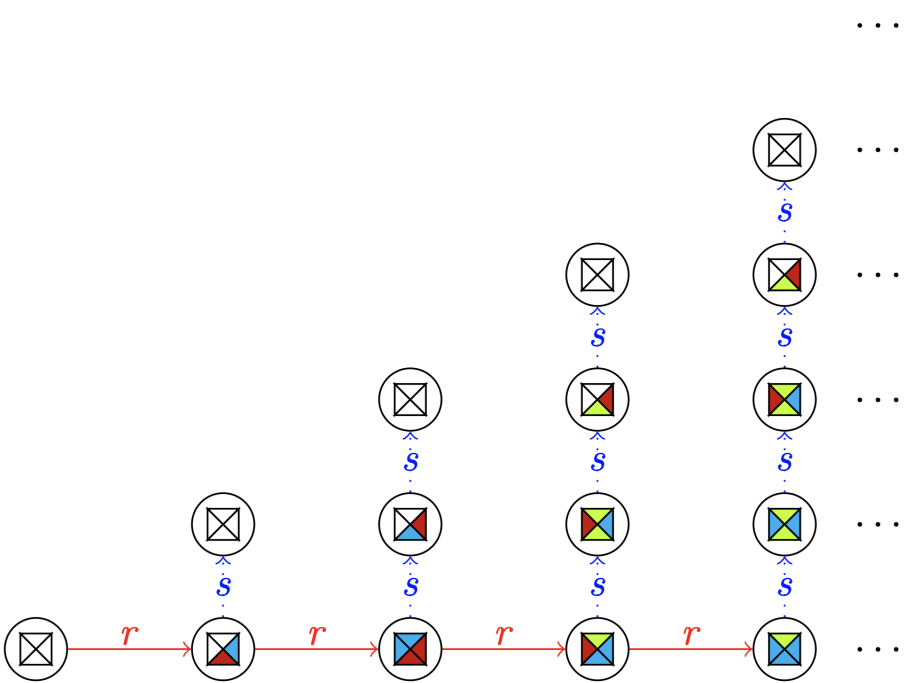
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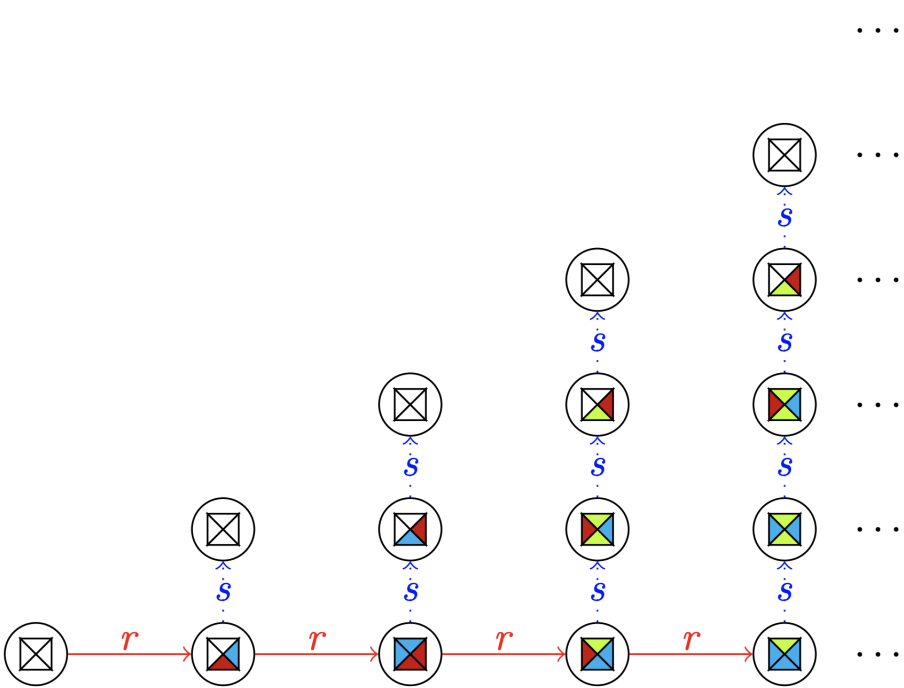

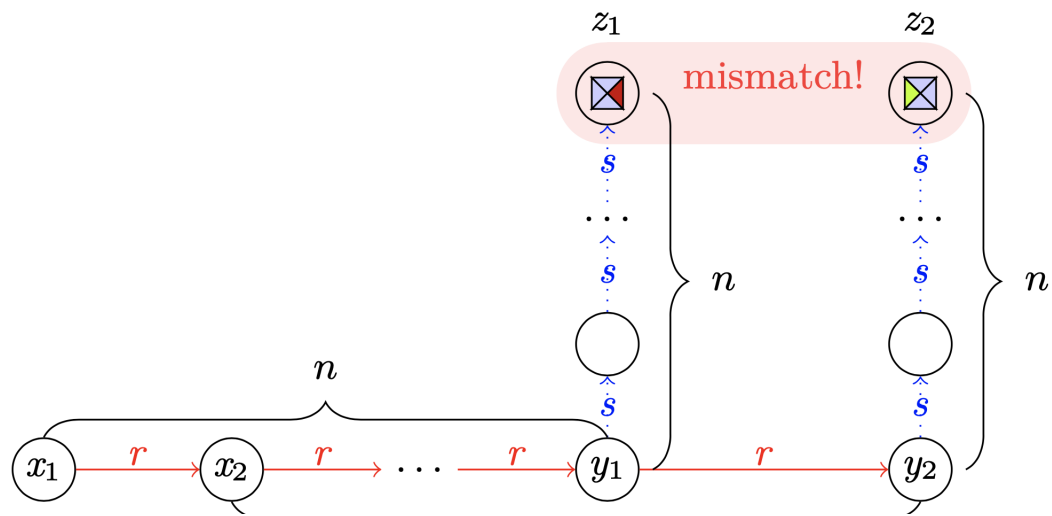

$$q^{\mathcal{D}}_{\blacktriangle} := \bigvee_{\text{t,t}' \text{ violating (OHori)}} \big[r(x_1,x_2) \wedge r^*(x_2,y_1) \wedge r(y_1,y_2) \wedge s^*(y_1,z_1) \wedge s^*(y_2,z_2) \wedge r^{\#} s^{\#}(x_1,z_1) \wedge r^{\#} s^{\#}(x_2,z_2) \wedge \mathrm{C}_{\mathrm{t}}(z_1) \wedge \mathrm{C}_{\mathrm{t}'}(z_2) \big]$$

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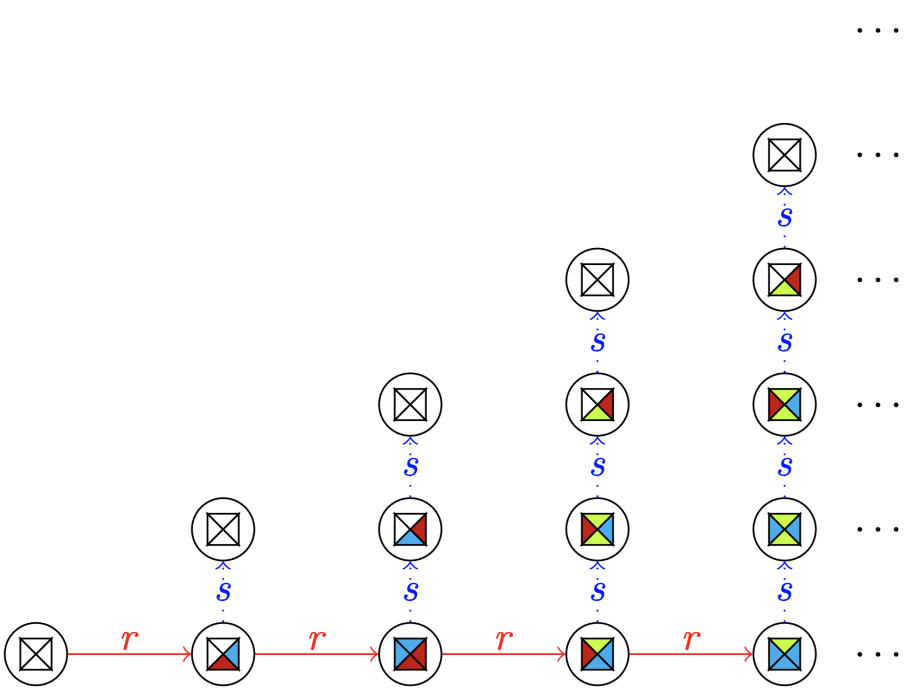

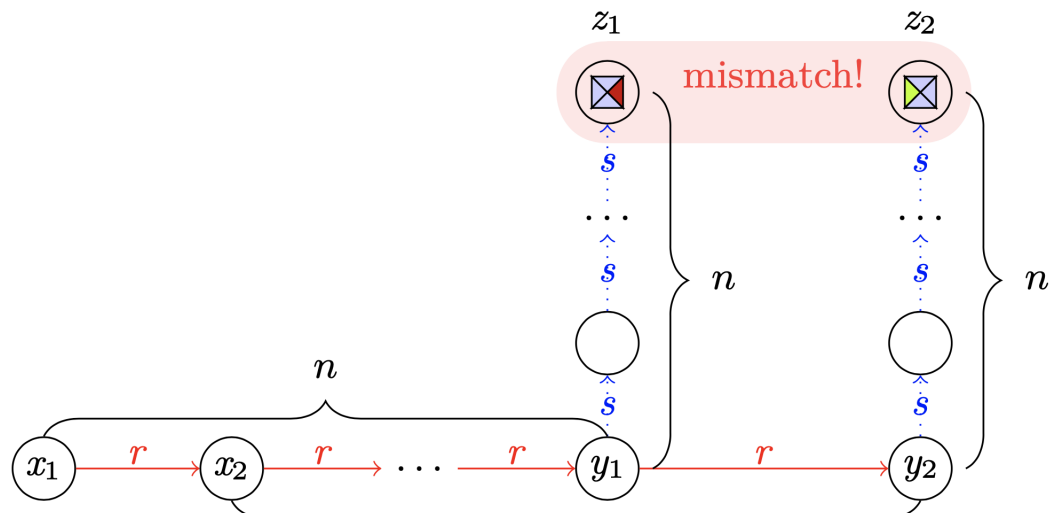

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Key Property: $\mathrm{C} \not\models q$ iff the octant can be covered.

Decidability of (extensions of) $\mathcal{ALC}_{\mathsf{reg}}$ do not transfer well to the non-regular setting.

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Loops

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Loops **Nominals**

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Loops

Vis. 1-counter

Nominals

Queries

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Looking for (postdoc?) job from Sept'24!

See: bartoszjanbednarczyk.github.io